



System Test Specification
for the
KNEAD Example System
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Balance Project

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DOCUMENT CHANGE HISTORY

The following table is a simple list of released revisions sent for review. Records of reviews and the review artifacts are saved with reviewer information in the The Balance Project artifact repository.

Change Record

Date	Version	Author(s)	Change Reference
13 Sep 2025	P1	Vinay Agarwal	Preliminary DRAFT version

Draft P1 Preliminary version of this document.

1. Baseline Document draft 1



TABLE OF CONTENTS

DOCUMENT CHANGE HISTORY	i
TABLE OF CONTENTS	ii
LIST OF TABLES	v
LIST OF FIGURES	vi
CHAPTER	
1 Scope	1
1.1 Identification	1
1.2 System Overview	1
1.3 Document Overview	1
1.3.1 Security and Privacy Considerations	1
1.3.2 Artifact Format	3
1.3.3 Document Version Information	4
2 References	5
2.1 Acronyms and Abbreviations	5
2.2 Glossary and Definitions	5
2.3 Referenced Documents	5
2.3.1 External Documents	6
2.3.2 Project Specific Documents	6
3 Test Preparations	7
3.1 Test1	7
3.1.1 Hardware Preparation	7
3.1.2 Software Preparation	7
3.1.3 Other Preparation	7
3.2 Test2	7
3.2.1 Hardware Preparation	7
3.2.2 Software Preparation	7
3.2.3 Other Preparation	7
3.3 Test3	7
3.3.1 Hardware Preparation	8
3.3.2 Software Preparation	8
3.3.3 Other Preparation	8
4 Test Specifications	9
4.1 Test1	9
4.1.1 Case1	9
4.1.1.1 Requirements Addressed	9
4.1.1.2 Prerequisite Conditions	9
4.1.1.3 Inputs	9
4.1.1.4 Expected Outputs	9



	CUI	System
	DISTRIBUTION RESTRICTIONS ON TITLE PAGE	Test Specification
4.1.1.5	Evaluation Criteria	9
4.1.1.6	Assumptions and Constraints	9
4.1.1.7	Procedure	9
4.1.2	Case2	13
4.1.2.1	Requirements Addressed	13
4.1.2.2	Prerequisite Conditions	13
4.1.2.3	Inputs	13
4.1.2.4	Expected Outputs	13
4.1.2.5	Evaluation Criteria	13
4.1.2.6	Assumptions and Constraints	14
4.1.2.7	Procedure	14
4.1.3	Case2	18
4.1.3.1	Requirements Addressed	18
4.1.3.2	Prerequisite Conditions	18
4.1.3.3	Inputs	18
4.1.3.4	Expected Outputs	18
4.1.3.5	Evaluation Criteria	18
4.1.3.6	Assumptions and Constraints	19
4.1.3.7	Procedure	19
4.2	Test2	23
4.2.1	Case1	23
4.2.1.1	Requirements Addressed	23
4.2.1.2	Prerequisite Conditions	23
4.2.1.3	Inputs	23
4.2.1.4	Expected Outputs	23
4.2.1.5	Evaluation Criteria	24
4.2.1.6	Assumptions and Constraints	24
4.2.1.7	Procedure	24
4.3	Test3	28
4.3.1	Case1	28
4.3.1.1	Requirements Addressed	28
4.3.1.2	Prerequisite Conditions	28
4.3.1.3	Inputs	28
4.3.1.4	Expected Outputs	28
4.3.1.5	Evaluation Criteria	29
4.3.1.6	Assumptions and Constraints	29
4.3.1.7	Procedure	29
4.3.2	Case2	33
4.3.2.1	Requirements Addressed	33
4.3.2.2	Prerequisite Conditions	33
4.3.2.3	Inputs	33
4.3.2.4	Expected Outputs	33
4.3.2.5	Evaluation Criteria	33
4.3.2.6	Assumptions and Constraints	34
4.3.2.7	Procedure	34

BALANCING PROJECT	iii CUI	KNEADSTS20240313-P1:128 Revision Date : 01 Dec, 2025
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4.3.3	Case3	38
4.3.3.1	Requirements Addressed	38
4.3.3.2	Prerequisite Conditions	38
4.3.3.3	Inputs	38
4.3.3.4	Expected Outputs	38
4.3.3.5	Evaluation Criteria	38
4.3.3.6	Assumptions and Constraints	39
4.3.3.7	Procedure	39
5	Traceability	44
6	Notes	45
6.1	Note Area 1	45
6.2	Note Area 2	45
APPENDIX		
Additional Information		46
Index		47



LIST OF TABLES

Table		Page
1	Acronym Definitions	5
2	Glossary Terms and Definitions	5



LIST OF FIGURES

Figure		Page
1	System Overview	2



CHAPTER 1

Scope

This document provides the System Test Specification (**STS**) for the Balance System. The system will be referred to as the Balance System.

1.1 Identification

The Balance System described in this document shall be known as Balance System version 1. However, the **STS** described herein shall be applicable to pre-releases such as Beta-releases for a phased release as listed for each requirement.

1.2 System Overview

Figure 1 shows the high-level architecture for the Balance System system. This diagram shows the major external interfaces that provide the capabilities of Balance System.

The Balance System is a single player game. The game is a standalone game played on the STM32 board. This system would be a game where the user would have to balance a ball on a LCD screen that is builtin on the STM32 board. The objective of the game is to balance the ball on the screen based on the way the board was tilted. Using an onboard gyroscope that communicates across the board to the MCU though SPI communication. Balance System would keep track of the current position of the ball and where the next updated move is. This helps keep track of the system of where the ball is until a movement has occurred. Balance System shall process at a maximum 180 Hz. This would give the user enough time to process the current angle of the ball and be able to present on the LCD-TFT screen.

1.3 Document Overview

This section provides information about this document's security/privacy considerations, contents, structure, and version information.

1.3.1 Security and Privacy Considerations

This document has been identified as "Controlled Unclassified Information" (CUI). Please follow the control block on the title page for ownership, creation, category, dissemination, and Point of Contact (**POC**) information. This information should be delineated, per <https://www.dodcui.mil> as:

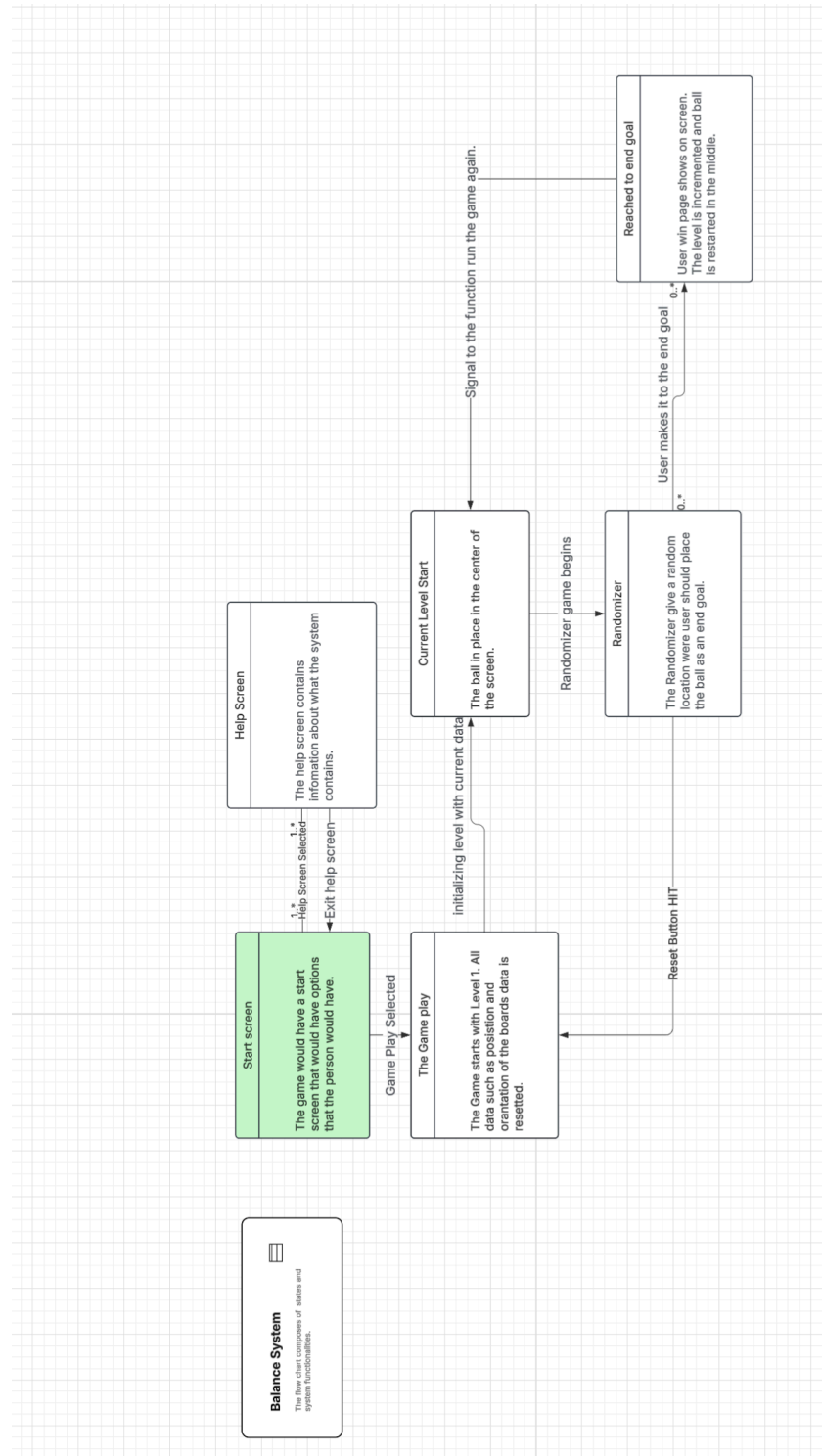


Figure 1: System Overview



Owner the name of the DoD Component (not required if identified in the letterhead)
Creator identification of the office creating the document
Category identification of the categories contained in the document
Dissemination applicable distribution statement or limited dissemination control (LDC)
POC name and phone number or email of POC

1.3.2 Artifact Format

This document format is based upon the guidance in the **STS DID** [ref`STD`DID] and the **STR DID** [ref`STR`DID]. The test planning is documented following the guidelines of ISO-12207 [ref`ISO`12207] and MIL-STD-498 [ref`MIL`STD`498] (from which ISO-12207 originated). This document follows the listed **STS** sub-section order.

Section 1 provides an overview of the system and this document.

Section 2 lists general and application-specific reference documents as well as glossary terms and acronyms.

Section 3 summarizes the test preparations.

Section 4 provides the detailed descriptions of the tests to be performed.¹

Section 5 provides any applicable requirement traceability.

Section 6 if needed, lists any general notes as may be applicable.

Appendices if needed, provide additional information as may be needed.

This document also is structured to serve as the basis for the system test report (**STR**). Each test is supplied with spaces for capturing pertinent hardware, software, and other log information. Each test is divided into one or more test cases, each with detailed steps, expected results for each step, and a set of easy to read pass/fail markers for each test step. All tests steps also provide space to fill in the results and to write notes and comments about each test step. The goal of this style is to generate this **STR** by scanning in the resultant **STS** with comments using this as the **STR** Appendix-A. In this manner, a “written” record of the testing is generated, thus saving money by not requiring a completely separate recording document for the **STR**.

¹This section follows the **DID** but places the test procedure details as the last section for each test case.



1.3.3 Document Version Information

This document was produced in $\text{\LaTeX}$ and *BibLaTeX/Biber*. The editing and document preparation were performed using $\text{\TeX}$ version 2.9 with the build option [$\text{\LaTeX} \Rightarrow \text{PS} \Rightarrow \text{PDF}$]. The $\text{\LaTeX}svn-multi$ package was used to glean SVN tracking information, when files are stored in an “SVN” version control system. The style `KNEADdocument` was used to provide the $\text{\LaTeX}$ and *BibLaTeX/Biber* formatting details.

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CHAPTER 2

References

This section provides a list of referenced items for this document.

2.1 Acronyms and Abbreviations

This section defines acronyms and abbreviations used in this and related documents.

Table 1: Acronym Definitions

Acronym	Definition
YSM	Yourdon Structured Method
LTDC	Layered Transmission Display Controller
STM	STMicroelectronics
MEMS	Micro-Electro-Mechanical Systems
End of acronym definition table	

2.2 Glossary and Definitions

This section defines glossary terms used in this and related documents.

Table 2: Glossary Terms and Definitions

Glossary Term	Definition
STM32F429I	Micro-controller board has all component fit onto one board.
Customer	The professor that is view the grading all assignments.
End of glossary terms table	

2.3 Referenced Documents

This section lists the referenced documents for this document. The references are categorized into two categories:



External Documents not directly associated with this project.

Project Documents that are directly associated with this project.

2.3.1 External Documents

2.3.2 Project Specific Documents



CHAPTER 3

Test Preparations

This section describes the test preparations to be performed for each test.

3.1 Test1

Test Test1 is to check the board loads properly.

3.1.1 Hardware Preparation

This hardware preparation for test Test1 is to have the board the the micro-usb connected to a power source.

3.1.2 Software Preparation

This software preparation for test Test1 is to have the current version of the code loaded to the board.

3.1.3 Other Preparation

This other preparation for test Test1 is not necessary.

3.2 Test2

Test Test2 is to check if the help screen would show when pressed.

3.2.1 Hardware Preparation

This hardware preparation for test Test2 is to have the board the the micro-usb connected to a power source.

3.2.2 Software Preparation

This software preparation for test Test2 is to have the current version of the code loaded to the board.

3.2.3 Other Preparation

This other preparation for test Test2 is not nessary.

3.3 Test3

Test Test3 is to check if the game screen would show when pressed.



3.3.1 Hardware Preparation

This hardware preparation for test Test3 is to have the board the the micro-usb connected to a power source.

3.3.2 Software Preparation

This software preparation for test Test3 is to have the current version of the code loaded to the board.

3.3.3 Other Preparation

This other preparation for test Test3 is not necessary.



CHAPTER 4

Test Specifications

This section specifies the tests to be performed.

4.1 Test1

Test Test1 is .

4.1.1 Case1

Test Test1 case Case1 is to see if the system turns on.

4.1.1.1 Requirements Addressed

This requirements validated by test Test1 case Case1 are listed in § 4.1.1.7.

4.1.1.2 Prerequisite Conditions

This prerequisite conditions for test Test1 case Case1 are that there must need to be at least 5 volts to be applied to the system.

4.1.1.3 Inputs

This inputs for test Test1 case Case1 are listed in § 4.1.1.7.

4.1.1.4 Expected Outputs

This expected outputs for test Test1 case Case1 are listed in § 4.1.1.7.

4.1.1.5 Evaluation Criteria

This evaluation criteria for test Test1 case Case1 are listed in § 4.1.1.7.

4.1.1.6 Assumptions and Constraints

This procedure for test Test1 case Case1 is not necessary.

4.1.1.7 Procedure

This procedure for test Test1 case Case1 is **...TBD....**

See step 4.3.3.7.1 -6 for how to reference specific steps.

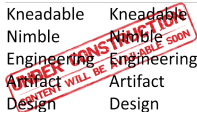
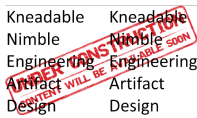


Test Procedure 4.1.1.7.1 Test Procedure 1	
Requirements Validated by Test Procedure 4.1.1.7.1	
R-4.1.1.7.1 -1	Fake Rqmt 1
R-4.1.1.7.1 -2	Fake Rqmt 2
Notes About Test Procedure 4.1.1.7.1	
N-4.1.1.7.1 -1	Note 1
N-4.1.1.7.1 -2	Note 2
Steps to be Performed for Test Procedure 4.1.1.7.1	
	START RECORDING OF PRE-TEST INFORMATION
S-4.1.1.7.1 -1	Record Date and Time at Start of Test
Expected Result	Date and Time at Start of Test are recorded.
PASS FAIL	Results :
S-4.1.1.7.1 -2	Record Name of Test Engineer(s) and Agency
Expected Result	Name of Test Engineer(s) and their Agency are recorded.
PASS FAIL	Results :
S-4.1.1.7.1 -3	Record Name of Witness(es) and Agency
Expected Result	Name of Witness(es) and their Agency are recorded.
PASS FAIL	Results :
S-4.1.1.7.1 -4	Record configuration information or name of file that contains such information.
Expected Result	Configuration information or name of file that contains such information is recorded.
PASS FAIL	Results :
	END RECORDING OF PRE-TEST INFORMATION
Test Procedure 4.1.1.7.1 – continues on the next page	



Test Procedure 4.1.1.7.1 – continued from previous page	
	CLOCK TEXT – good to show that some time must elapse
	HAND TEXT – use, with text, when tester needs to pause and double check things
	INFO TEXT – good to provide information needed at this point in the test
	KEY TEXT – good to make a key point, or if something needs to be locked/unlocked
	MAGNIFY TEXT – note info that magnifies what is happening
	PLAYARROW TEXT – denote a starting point, such as when test stations change. This is just more text to see what happens when there are 3 or 4 lines of text w.r.t. centering of icon. These text blocks should be short, but, could be long, so this checks to see what happens with 5 or 6 lines of text.
	BANG TEXT – denote a WARNING
	SHOCK TEXT – denote a HAZARD
	RADIATION TEXT – denote an EXTREME HAZARD
	ON/OFF TEXT – denote a POWER ON or OFF action
	FOOTSTEPS TEXT – denote a MOVEMENT action
	REPEAT TEXT – denote a repeat cycle or action
	QUESTION TEXT – ensure a question is answered
Test Procedure 4.1.1.7.1 – continues on the next page	



Test Procedure 4.1.1.7.1 – continued from previous page	
	BULLSEYE TEXT – denote the end of a mini-sequence
	FILES TEXT – denote storage of data or records
S-4.1.1.7.1 -5	Do Step 1
Expected Result	Step 1 works!
PASS FAIL	Results :
S-4.1.1.7.1 -6	Do Labeled Step 2
Expected Result	Step 2 works!
PASS FAIL	Results :
S-4.1.1.7.1 -7	Do stuff so the system looks like the expected image below.
Expected Result	
PASS FAIL	Results :
S-4.1.1.7.1 -8	
Expected Result	Labeled step 4 works!
PASS FAIL	Results :
	START RECORDING OF POST-TEST INFORMATION
S-4.1.1.7.1 -9	Record Date and Time at End of Test
Expected Result	Date and Time at End of Test are recorded.
PASS FAIL	Results :
S-4.1.1.7.1 -10	Record Signature of Test Engineer(s)
Expected Result	Signature of Test Engineer(s) is recorded.
PASS FAIL	Results :
Test Procedure 4.1.1.7.1 – continues on the next page	



Test Procedure 4.1.1.7.1 – continued from previous page	
S-4.1.1.7.1 -11	Record Signature of Witness(es)
Expected Result	Signature of Witness(es) are recorded.
PASS FAIL	Results :
S-4.1.1.7.1 -12	Record any pertinent comment about the test procedure, results, and/or environment.
Expected Result	Any pertinent comment about the test procedure, results, and/or environment is recorded.
PASS FAIL	Results :
	END RECORDING OF POST-TEST INFORMATION
End of Test Procedure 4.1.1.7.1	

4.1.2 Case2

Test Test1 case Case2 is the help screen showing up.

4.1.2.1 Requirements Addressed

This requirements validated by test Test1 case Case2 are listed in § 4.1.3.7.

4.1.2.2 Prerequisite Conditions

This prerequisite conditions for test Test1 case Case2 are not necessary.

4.1.2.3 Inputs

This inputs for test Test1 case Case2 are listed in § 4.1.3.7.

4.1.2.4 Expected Outputs

This expected outputs for test Test1 case Case2 are listed in § 4.1.3.7.

4.1.2.5 Evaluation Criteria

This evaluation criteria for test Test1 case Case2 are listed in § 4.1.3.7.



4.1.2.6 Assumptions and Constraints

This procedure for test Test1 case Case2 is not necessary.

4.1.2.7 Procedure

This procedure for test Test1 case Case2 is not necessary.

See step 4.3.3.7.1 -6 for how to reference specific steps.

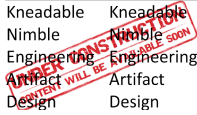
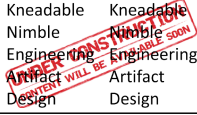


Test Procedure 4.1.2.7.1	
Test Procedure 1	
Requirements Validated by Test Procedure 4.1.2.7.1	
R-4.1.2.7.1 -1	Fake Rqmt 1
R-4.1.2.7.1 -2	Fake Rqmt 2
Notes About Test Procedure 4.1.2.7.1	
N-4.1.2.7.1 -1	Note 1
N-4.1.2.7.1 -2	Note 2
Steps to be Performed for Test Procedure 4.1.2.7.1	
	START RECORDING OF PRE-TEST INFORMATION
S-4.1.2.7.1 -1	Record Date and Time at Start of Test
Expected Result	Date and Time at Start of Test are recorded.
PASS FAIL	Results :
S-4.1.2.7.1 -2	Record Name of Test Engineer(s) and Agency
Expected Result	Name of Test Engineer(s) and their Agency are recorded.
PASS FAIL	Results :
S-4.1.2.7.1 -3	Record Name of Witness(es) and Agency
Expected Result	Name of Witness(es) and their Agency are recorded.
PASS FAIL	Results :
S-4.1.2.7.1 -4	Record configuration information or name of file that contains such information.
Expected Result	Configuration information or name of file that contains such information is recorded.
PASS FAIL	Results :
	END RECORDING OF PRE-TEST INFORMATION
Test Procedure 4.1.2.7.1 – continues on the next page	



Test Procedure 4.1.2.7.1 – continued from previous page	
	CLOCK TEXT – good to show that some time must elapse
	HAND TEXT – use, with text, when tester needs to pause and double check things
	INFO TEXT – good to provide information needed at this point in the test
	KEY TEXT – good to make a key point, or if something needs to be locked/unlocked
	MAGNIFY TEXT – note info that magnifies what is happening
	PLAYARROW TEXT – denote a starting point, such as when test stations change. This is just more text to see what happens when there are 3 or 4 lines of text w.r.t. centering of icon. These text blocks should be short, but, could be long, so this checks to see what happens with 5 or 6 lines of text.
	BANG TEXT – denote a WARNING
	SHOCK TEXT – denote a HAZARD
	RADIATION TEXT – denote an EXTREME HAZARD
	ON/OFF TEXT – denote a POWER ON or OFF action
	FOOTSTEPS TEXT – denote a MOVEMENT action
	REPEAT TEXT – denote a repeat cycle or action
	QUESTION TEXT – ensure a question is answered
Test Procedure 4.1.2.7.1 – continues on the next page	



Test Procedure 4.1.2.7.1 – continued from previous page	
	BULLSEYE TEXT – denote the end of a mini-sequence
	FILES TEXT – denote storage of data or records
S-4.1.2.7.1 -5	Do Step 1
Expected Result	Step 1 works!
PASS FAIL	Results :
S-4.1.2.7.1 -6	Do Labeled Step 2
Expected Result	Step 2 works!
PASS FAIL	Results :
S-4.1.2.7.1 -7	Do stuff so the system looks like the expected image below.
Expected Result	
PASS FAIL	Results :
S-4.1.2.7.1 -8	
Expected Result	Labeled step 4 works!
PASS FAIL	Results :
	START RECORDING OF POST-TEST INFORMATION
S-4.1.2.7.1 -9	Record Date and Time at End of Test
Expected Result	Date and Time at End of Test are recorded.
PASS FAIL	Results :
S-4.1.2.7.1 -10	Record Signature of Test Engineer(s)
Expected Result	Signature of Test Engineer(s) is recorded.
PASS FAIL	Results :
Test Procedure 4.1.2.7.1 – continues on the next page	



Test Procedure 4.1.2.7.1 – continued from previous page	
S-4.1.2.7.1 -11	Record Signature of Witness(es)
Expected Result	Signature of Witness(es) are recorded.
PASS FAIL	Results :
S-4.1.2.7.1 -12	Record any pertinent comment about the test procedure, results, and/or environment.
Expected Result	Any pertinent comment about the test procedure, results, and/or environment is recorded.
PASS FAIL	Results :
	END RECORDING OF POST-TEST INFORMATION
End of Test Procedure 4.1.2.7.1	

4.1.3 Case2

Test Test1 case Case2 is the game working.

4.1.3.1 Requirements Addressed

This requirements validated by test Test1 case Case2 are listed in § 4.1.3.7.

4.1.3.2 Prerequisite Conditions

This prerequisite conditions for test Test1 case Case2 are not necessary.

4.1.3.3 Inputs

This inputs for test Test1 case Case2 are listed in § 4.1.3.7.

4.1.3.4 Expected Outputs

This expected outputs for test Test1 case Case2 are listed in § 4.1.3.7.

4.1.3.5 Evaluation Criteria

This evaluation criteria for test Test1 case Case2 are listed in § 4.1.3.7.



4.1.3.6 Assumptions and Constraints

This procedure for test Test1 case Case2 is not necessary.

4.1.3.7 Procedure

This procedure for test Test1 case Case2 is not necessary.

See step 4.3.3.7.1 -6 for how to reference specific steps.

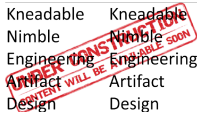
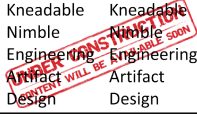


Test Procedure 4.1.3.7.1 Test Procedure 1	
Requirements Validated by Test Procedure 4.1.3.7.1	
R-4.1.3.7.1 -1	Fake Rqmt 1
R-4.1.3.7.1 -2	Fake Rqmt 2
Notes About Test Procedure 4.1.3.7.1	
N-4.1.3.7.1 -1	Note 1
N-4.1.3.7.1 -2	Note 2
Steps to be Performed for Test Procedure 4.1.3.7.1	
	START RECORDING OF PRE-TEST INFORMATION
S-4.1.3.7.1 -1	Record Date and Time at Start of Test
Expected Result	Date and Time at Start of Test are recorded.
PASS FAIL	Results :
S-4.1.3.7.1 -2	Record Name of Test Engineer(s) and Agency
Expected Result	Name of Test Engineer(s) and their Agency are recorded.
PASS FAIL	Results :
S-4.1.3.7.1 -3	Record Name of Witness(es) and Agency
Expected Result	Name of Witness(es) and their Agency are recorded.
PASS FAIL	Results :
S-4.1.3.7.1 -4	Record configuration information or name of file that contains such information.
Expected Result	Configuration information or name of file that contains such information is recorded.
PASS FAIL	Results :
	END RECORDING OF PRE-TEST INFORMATION
Test Procedure 4.1.3.7.1 – continues on the next page	



Test Procedure 4.1.3.7.1 – continued from previous page	
	CLOCK TEXT – good to show that some time must elapse
	HAND TEXT – use, with text, when tester needs to pause and double check things
	INFO TEXT – good to provide information needed at this point in the test
	KEY TEXT – good to make a key point, or if something needs to be locked/unlocked
	MAGNIFY TEXT – note info that magnifies what is happening
	PLAYARROW TEXT – denote a starting point, such as when test stations change. This is just more text to see what happens when there are 3 or 4 lines of text w.r.t. centering of icon. These text blocks should be short, but, could be long, so this checks to see what happens with 5 or 6 lines of text.
	BANG TEXT – denote a WARNING
	SHOCK TEXT – denote a HAZARD
	RADIATION TEXT – denote an EXTREME HAZARD
	ON/OFF TEXT – denote a POWER ON or OFF action
	FOOTSTEPS TEXT – denote a MOVEMENT action
	REPEAT TEXT – denote a repeat cycle or action
	QUESTION TEXT – ensure a question is answered
Test Procedure 4.1.3.7.1 – continues on the next page	



Test Procedure 4.1.3.7.1 – continued from previous page	
	BULLSEYE TEXT – denote the end of a mini-sequence
	FILES TEXT – denote storage of data or records
S-4.1.3.7.1 -5	Do Step 1
Expected Result	Step 1 works!
PASS FAIL	Results :
S-4.1.3.7.1 -6	Do Labeled Step 2
Expected Result	Step 2 works!
PASS FAIL	Results :
S-4.1.3.7.1 -7	Do stuff so the system looks like the expected image below.
Expected Result	
PASS FAIL	Results :
S-4.1.3.7.1 -8	
Expected Result	Labeled step 4 works!
PASS FAIL	Results :
	START RECORDING OF POST-TEST INFORMATION
S-4.1.3.7.1 -9	Record Date and Time at End of Test
Expected Result	Date and Time at End of Test are recorded.
PASS FAIL	Results :
S-4.1.3.7.1 -10	Record Signature of Test Engineer(s)
Expected Result	Signature of Test Engineer(s) is recorded.
PASS FAIL	Results :
Test Procedure 4.1.3.7.1 – continues on the next page	



Test Procedure 4.1.3.7.1 – continued from previous page	
S-4.1.3.7.1 -11	Record Signature of Witness(es)
Expected Result	Signature of Witness(es) are recorded.
PASS FAIL	Results :
S-4.1.3.7.1 -12	Record any pertinent comment about the test procedure, results, and/or environment.
Expected Result	Any pertinent comment about the test procedure, results, and/or environment is recorded.
PASS FAIL	Results :
	END RECORDING OF POST-TEST INFORMATION
End of Test Procedure 4.1.3.7.1	

4.2 Test2

Test Test2 is .

4.2.1 Case1

Test Test2 case Case1 is the help screen showing up.

4.2.1.1 Requirements Addressed

This requirements validated by test Test2 case Case1 are listed in § 4.2.1.7.

4.2.1.2 Prerequisite Conditions

This prerequisite conditions for test Test2 case Case1 are not necessary.

4.2.1.3 Inputs

This inputs for test Test2 case Case1 are listed in § 4.2.1.7.

4.2.1.4 Expected Outputs

This expected outputs for test Test2 case Case1 are listed in § 4.2.1.7.



4.2.1.5 Evaluation Criteria

This evaluation criteria for test Test2 case Case1 are listed in § 4.2.1.7.

4.2.1.6 Assumptions and Constraints

This procedure for test Test2 case Case1 is not necessary.

4.2.1.7 Procedure

This procedure for test Test2 case Case1 is not necessary.

See step 4.3.3.7.1 -6 for how to reference specific steps.

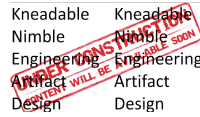
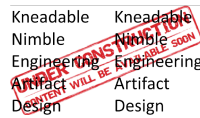


Test Procedure 4.2.1.7.1 Test Procedure 1	
Requirements Validated by Test Procedure 4.2.1.7.1	
R-4.2.1.7.1 -1	Fake Rqmt 1
R-4.2.1.7.1 -2	Fake Rqmt 2
Notes About Test Procedure 4.2.1.7.1	
N-4.2.1.7.1 -1	Note 1
N-4.2.1.7.1 -2	Note 2
Steps to be Performed for Test Procedure 4.2.1.7.1	
	START RECORDING OF PRE-TEST INFORMATION
S-4.2.1.7.1 -1	Record Date and Time at Start of Test
Expected Result	Date and Time at Start of Test are recorded.
PASS FAIL	Results :
S-4.2.1.7.1 -2	Record Name of Test Engineer(s) and Agency
Expected Result	Name of Test Engineer(s) and their Agency are recorded.
PASS FAIL	Results :
S-4.2.1.7.1 -3	Record Name of Witness(es) and Agency
Expected Result	Name of Witness(es) and their Agency are recorded.
PASS FAIL	Results :
S-4.2.1.7.1 -4	Record configuration information or name of file that contains such information.
Expected Result	Configuration information or name of file that contains such information is recorded.
PASS FAIL	Results :
	END RECORDING OF PRE-TEST INFORMATION
Test Procedure 4.2.1.7.1 – continues on the next page	



Test Procedure 4.2.1.7.1 – continued from previous page	
	CLOCK TEXT – good to show that some time must elapse
	HAND TEXT – use, with text, when tester needs to pause and double check things
	INFO TEXT – good to provide information needed at this point in the test
	KEY TEXT – good to make a key point, or if something needs to be locked/unlocked
	MAGNIFY TEXT – note info that magnifies what is happening
	PLAYARROW TEXT – denote a starting point, such as when test stations change. This is just more text to see what happens when there are 3 or 4 lines of text w.r.t. centering of icon. These text blocks should be short, but, could be long, so this checks to see what happens with 5 or 6 lines of text.
	BANG TEXT – denote a WARNING
	SHOCK TEXT – denote a HAZARD
	RADIATION TEXT – denote an EXTREME HAZARD
	ON/OFF TEXT – denote a POWER ON or OFF action
	FOOTSTEPS TEXT – denote a MOVEMENT action
	REPEAT TEXT – denote a repeat cycle or action
	QUESTION TEXT – ensure a question is answered
Test Procedure 4.2.1.7.1 – continues on the next page	



Test Procedure 4.2.1.7.1 – continued from previous page	
	BULLSEYE TEXT – denote the end of a mini-sequence
	FILES TEXT – denote storage of data or records
S-4.2.1.7.1 -5	Do Step 1
Expected Result	Step 1 works!
PASS FAIL	Results :
S-4.2.1.7.1 -6	Do Labeled Step 2
Expected Result	Step 2 works!
PASS FAIL	Results :
S-4.2.1.7.1 -7	Do stuff so the system looks like the expected image below.
Expected Result	
PASS FAIL	Results :
S-4.2.1.7.1 -8	
Expected Result	Labeled step 4 works!
PASS FAIL	Results :
	START RECORDING OF POST-TEST INFORMATION
S-4.2.1.7.1 -9	Record Date and Time at End of Test
Expected Result	Date and Time at End of Test are recorded.
PASS FAIL	Results :
S-4.2.1.7.1 -10	Record Signature of Test Engineer(s)
Expected Result	Signature of Test Engineer(s) is recorded.
PASS FAIL	Results :
Test Procedure 4.2.1.7.1 – continues on the next page	



Test Procedure 4.2.1.7.1 – continued from previous page	
S-4.2.1.7.1 -11	Record Signature of Witness(es)
Expected Result	Signature of Witness(es) are recorded.
PASS FAIL	Results :
S-4.2.1.7.1 -12	Record any pertinent comment about the test procedure, results, and/or environment.
Expected Result	Any pertinent comment about the test procedure, results, and/or environment is recorded.
PASS FAIL	Results :
	END RECORDING OF POST-TEST INFORMATION
End of Test Procedure 4.2.1.7.1	

4.3 Test3

Test Test3 is a system level test.

4.3.1 Case1

Test Test3 case Case1 is the game working.

4.3.1.1 Requirements Addressed

This requirements validated by test Test3 case Case1 are listed in § 4.3.1.7.

4.3.1.2 Prerequisite Conditions

This prerequisite conditions for test Test3 case Case1 are not necessary.

4.3.1.3 Inputs

This inputs for test Test3 case Case1 are listed in § 4.3.1.7.

4.3.1.4 Expected Outputs

This expected outputs for test Test3 case Case1 are listed in § 4.3.1.7.



4.3.1.5 Evaluation Criteria

This evaluation criteria for test Test3 case Case1 are listed in § 4.3.1.7.

4.3.1.6 Assumptions and Constraints

This procedure for test Test3 case Case1 is not necessary.

4.3.1.7 Procedure

This procedure for test Test3 case Case1 is not necessary.

See step 4.3.3.7.1 -6 for how to reference specific steps.

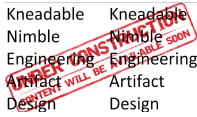
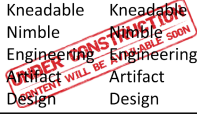


Test Procedure 4.3.1.7.1	
Test Procedure 1	
Requirements Validated by Test Procedure 4.3.1.7.1	
R-4.3.1.7.1 -1	Fake Rqmt 1
R-4.3.1.7.1 -2	Fake Rqmt 2
Notes About Test Procedure 4.3.1.7.1	
N-4.3.1.7.1 -1	Note 1
N-4.3.1.7.1 -2	Note 2
Steps to be Performed for Test Procedure 4.3.1.7.1	
	START RECORDING OF PRE-TEST INFORMATION
S-4.3.1.7.1 -1	Record Date and Time at Start of Test
Expected Result	Date and Time at Start of Test are recorded.
PASS FAIL	Results :
S-4.3.1.7.1 -2	Record Name of Test Engineer(s) and Agency
Expected Result	Name of Test Engineer(s) and their Agency are recorded.
PASS FAIL	Results :
S-4.3.1.7.1 -3	Record Name of Witness(es) and Agency
Expected Result	Name of Witness(es) and their Agency are recorded.
PASS FAIL	Results :
S-4.3.1.7.1 -4	Record configuration information or name of file that contains such information.
Expected Result	Configuration information or name of file that contains such information is recorded.
PASS FAIL	Results :
	END RECORDING OF PRE-TEST INFORMATION
Test Procedure 4.3.1.7.1 – continues on the next page	



Test Procedure 4.3.1.7.1 – continued from previous page	
	CLOCK TEXT – good to show that some time must elapse
	HAND TEXT – use, with text, when tester needs to pause and double check things
	INFO TEXT – good to provide information needed at this point in the test
	KEY TEXT – good to make a key point, or if something needs to be locked/unlocked
	MAGNIFY TEXT – note info that magnifies what is happening
	PLAYARROW TEXT – denote a starting point, such as when test stations change. This is just more text to see what happens when there are 3 or 4 lines of text w.r.t. centering of icon. These text blocks should be short, but, could be long, so this checks to see what happens with 5 or 6 lines of text.
	BANG TEXT – denote a WARNING
	SHOCK TEXT – denote a HAZARD
	RADIATION TEXT – denote an EXTREME HAZARD
	ON/OFF TEXT – denote a POWER ON or OFF action
	FOOTSTEPS TEXT – denote a MOVEMENT action
	REPEAT TEXT – denote a repeat cycle or action
	QUESTION TEXT – ensure a question is answered
Test Procedure 4.3.1.7.1 – continues on the next page	



Test Procedure 4.3.1.7.1 – continued from previous page	
	BULLSEYE TEXT – denote the end of a mini-sequence
	FILES TEXT – denote storage of data or records
S-4.3.1.7.1 -5	Do Step 1
Expected Result	Step 1 works!
PASS FAIL	Results :
S-4.3.1.7.1 -6	Do Labeled Step 2
Expected Result	Step 2 works!
PASS FAIL	Results :
S-4.3.1.7.1 -7	Do stuff so the system looks like the expected image below.
Expected Result	
PASS FAIL	Results :
S-4.3.1.7.1 -8	
Expected Result	Labeled step 4 works!
PASS FAIL	Results :
	START RECORDING OF POST-TEST INFORMATION
S-4.3.1.7.1 -9	Record Date and Time at End of Test
Expected Result	Date and Time at End of Test are recorded.
PASS FAIL	Results :
S-4.3.1.7.1 -10	Record Signature of Test Engineer(s)
Expected Result	Signature of Test Engineer(s) is recorded.
PASS FAIL	Results :
Test Procedure 4.3.1.7.1 – continues on the next page	



Test Procedure 4.3.1.7.1 – continued from previous page	
S-4.3.1.7.1 -11	Record Signature of Witness(es)
Expected Result	Signature of Witness(es) are recorded.
PASS FAIL	Results :
S-4.3.1.7.1 -12	Record any pertinent comment about the test procedure, results, and/or environment.
Expected Result	Any pertinent comment about the test procedure, results, and/or environment is recorded.
PASS FAIL	Results :
	END RECORDING OF POST-TEST INFORMATION
End of Test Procedure 4.3.1.7.1	

4.3.2 Case2

Test Test3 case Case2 is the help screen showing up.

4.3.2.1 Requirements Addressed

This requirements validated by test Test3 case Case2 are listed in § 4.3.2.7.

4.3.2.2 Prerequisite Conditions

This prerequisite conditions for test Test3 case Case2 are not necessary.

4.3.2.3 Inputs

This inputs for test Test3 case Case2 are listed in § 4.3.2.7.

4.3.2.4 Expected Outputs

This expected outputs for test Test3 case Case2 are listed in § 4.3.2.7.

4.3.2.5 Evaluation Criteria

This evaluation criteria for test Test3 case Case2 are listed in § 4.3.2.7.



4.3.2.6 Assumptions and Constraints

This procedure for test Test3 case Case2 is not necessary.

4.3.2.7 Procedure

This procedure for test Test3 case Case2 is not necessary.

See step 4.3.3.7.1 -6 for how to reference specific steps.

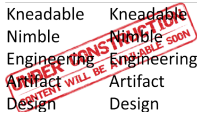
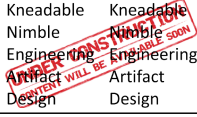


Test Procedure 4.3.2.7.1	
Test Procedure 1	
Requirements Validated by Test Procedure 4.3.2.7.1	
R-4.3.2.7.1 -1	Fake Rqmt 1
R-4.3.2.7.1 -2	Fake Rqmt 2
Notes About Test Procedure 4.3.2.7.1	
N-4.3.2.7.1 -1	Note 1
N-4.3.2.7.1 -2	Note 2
Steps to be Performed for Test Procedure 4.3.2.7.1	
	START RECORDING OF PRE-TEST INFORMATION
S-4.3.2.7.1 -1	Record Date and Time at Start of Test
Expected Result	Date and Time at Start of Test are recorded.
PASS FAIL	Results :
S-4.3.2.7.1 -2	Record Name of Test Engineer(s) and Agency
Expected Result	Name of Test Engineer(s) and their Agency are recorded.
PASS FAIL	Results :
S-4.3.2.7.1 -3	Record Name of Witness(es) and Agency
Expected Result	Name of Witness(es) and their Agency are recorded.
PASS FAIL	Results :
S-4.3.2.7.1 -4	Record configuration information or name of file that contains such information.
Expected Result	Configuration information or name of file that contains such information is recorded.
PASS FAIL	Results :
	END RECORDING OF PRE-TEST INFORMATION
Test Procedure 4.3.2.7.1 – continues on the next page	



Test Procedure 4.3.2.7.1 – continued from previous page	
	CLOCK TEXT – good to show that some time must elapse
	HAND TEXT – use, with text, when tester needs to pause and double check things
	INFO TEXT – good to provide information needed at this point in the test
	KEY TEXT – good to make a key point, or if something needs to be locked/unlocked
	MAGNIFY TEXT – note info that magnifies what is happening
	PLAYARROW TEXT – denote a starting point, such as when test stations change. This is just more text to see what happens when there are 3 or 4 lines of text w.r.t. centering of icon. These text blocks should be short, but, could be long, so this checks to see what happens with 5 or 6 lines of text.
	BANG TEXT – denote a WARNING
	SHOCK TEXT – denote a HAZARD
	RADIATION TEXT – denote an EXTREME HAZARD
	ON/OFF TEXT – denote a POWER ON or OFF action
	FOOTSTEPS TEXT – denote a MOVEMENT action
	REPEAT TEXT – denote a repeat cycle or action
	QUESTION TEXT – ensure a question is answered
Test Procedure 4.3.2.7.1 – continues on the next page	



Test Procedure 4.3.2.7.1 – continued from previous page	
	BULLSEYE TEXT – denote the end of a mini-sequence
	FILES TEXT – denote storage of data or records
S-4.3.2.7.1 -5	Do Step 1
Expected Result	Step 1 works!
PASS FAIL	Results :
S-4.3.2.7.1 -6	Do Labeled Step 2
Expected Result	Step 2 works!
PASS FAIL	Results :
S-4.3.2.7.1 -7	Do stuff so the system looks like the expected image below.
Expected Result	
PASS FAIL	Results :
S-4.3.2.7.1 -8	
Expected Result	Labeled step 4 works!
PASS FAIL	Results :
	START RECORDING OF POST-TEST INFORMATION
S-4.3.2.7.1 -9	Record Date and Time at End of Test
Expected Result	Date and Time at End of Test are recorded.
PASS FAIL	Results :
S-4.3.2.7.1 -10	Record Signature of Test Engineer(s)
Expected Result	Signature of Test Engineer(s) is recorded.
PASS FAIL	Results :
Test Procedure 4.3.2.7.1 – continues on the next page	



Test Procedure 4.3.2.7.1 – continued from previous page	
S-4.3.2.7.1 -11	Record Signature of Witness(es)
Expected Result	Signature of Witness(es) are recorded.
PASS FAIL	Results :
S-4.3.2.7.1 -12	Record any pertinent comment about the test procedure, results, and/or environment.
Expected Result	Any pertinent comment about the test procedure, results, and/or environment is recorded.
PASS FAIL	Results :
	END RECORDING OF POST-TEST INFORMATION
End of Test Procedure 4.3.2.7.1	

4.3.3 Case3

Test Test3 case Case3 is the game working.

4.3.3.1 Requirements Addressed

This requirements validated by test Test3 case Case3 are listed in § 4.3.3.7.

4.3.3.2 Prerequisite Conditions

This prerequisite conditions for test Test3 case Case3 are not necessary.

4.3.3.3 Inputs

This inputs for test Test3 case Case3 are listed in § 4.3.3.7.

4.3.3.4 Expected Outputs

This expected outputs for test Test3 case Case3 are listed in § 4.3.3.7.

4.3.3.5 Evaluation Criteria

This evaluation criteria for test Test3 case Case3 are listed in § 4.3.3.7.



4.3.3.6 Assumptions and Constraints

This procedure for test Test3 case Case3 is not necessary.

4.3.3.7 Procedure

This procedure for test Test3 case Case3 is not necessary.

See step 4.3.3.7.1 -6 for how to reference specific steps.

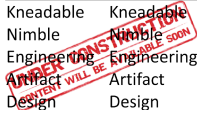
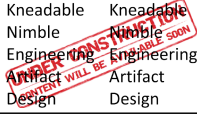


Test Procedure 4.3.3.7.1	
Test Procedure 1	
Requirements Validated by Test Procedure 4.3.3.7.1	
R-4.3.3.7.1 -1	Fake Rqmt 1
R-4.3.3.7.1 -2	Fake Rqmt 2
Notes About Test Procedure 4.3.3.7.1	
N-4.3.3.7.1 -1	Note 1
N-4.3.3.7.1 -2	Note 2
Steps to be Performed for Test Procedure 4.3.3.7.1	
	START RECORDING OF PRE-TEST INFORMATION
S-4.3.3.7.1 -1	Record Date and Time at Start of Test
Expected Result	Date and Time at Start of Test are recorded.
PASS FAIL	Results :
S-4.3.3.7.1 -2	Record Name of Test Engineer(s) and Agency
Expected Result	Name of Test Engineer(s) and their Agency are recorded.
PASS FAIL	Results :
S-4.3.3.7.1 -3	Record Name of Witness(es) and Agency
Expected Result	Name of Witness(es) and their Agency are recorded.
PASS FAIL	Results :
S-4.3.3.7.1 -4	Record configuration information or name of file that contains such information.
Expected Result	Configuration information or name of file that contains such information is recorded.
PASS FAIL	Results :
	END RECORDING OF PRE-TEST INFORMATION
Test Procedure 4.3.3.7.1 – continues on the next page	



Test Procedure 4.3.3.7.1 – continued from previous page	
	CLOCK TEXT – good to show that some time must elapse
	HAND TEXT – use, with text, when tester needs to pause and double check things
	INFO TEXT – good to provide information needed at this point in the test
	KEY TEXT – good to make a key point, or if something needs to be locked/unlocked
	MAGNIFY TEXT – note info that magnifies what is happening
	PLAYARROW TEXT – denote a starting point, such as when test stations change. This is just more text to see what happens when there are 3 or 4 lines of text w.r.t. centering of icon. These text blocks should be short, but, could be long, so this checks to see what happens with 5 or 6 lines of text.
	BANG TEXT – denote a WARNING
	SHOCK TEXT – denote a HAZARD
	RADIATION TEXT – denote an EXTREME HAZARD
	ON/OFF TEXT – denote a POWER ON or OFF action
	FOOTSTEPS TEXT – denote a MOVEMENT action
	REPEAT TEXT – denote a repeat cycle or action
	QUESTION TEXT – ensure a question is answered
Test Procedure 4.3.3.7.1 – continues on the next page	



Test Procedure 4.3.3.7.1 – continued from previous page	
	BULLSEYE TEXT – denote the end of a mini-sequence
	FILES TEXT – denote storage of data or records
S-4.3.3.7.1 -5	Do Step 1
Expected Result	Step 1 works!
PASS FAIL	Results :
S-4.3.3.7.1 -6	Do Labeled Step 2
Expected Result	Step 2 works!
PASS FAIL	Results :
S-4.3.3.7.1 -7	Do stuff so the system looks like the expected image below.
Expected Result	
PASS FAIL	Results :
S-4.3.3.7.1 -8	
Expected Result	Labeled step 4 works!
PASS FAIL	Results :
	START RECORDING OF POST-TEST INFORMATION
S-4.3.3.7.1 -9	Record Date and Time at End of Test
Expected Result	Date and Time at End of Test are recorded.
PASS FAIL	Results :
S-4.3.3.7.1 -10	Record Signature of Test Engineer(s)
Expected Result	Signature of Test Engineer(s) is recorded.
PASS FAIL	Results :
Test Procedure 4.3.3.7.1 – continues on the next page	



Test Procedure 4.3.3.7.1 – continued from previous page	
S-4.3.3.7.1 -11	Record Signature of Witness(es)
Expected Result	Signature of Witness(es) are recorded.
PASS FAIL	Results :
S-4.3.3.7.1 -12	Record any pertinent comment about the test procedure, results, and/or environment.
Expected Result	Any pertinent comment about the test procedure, results, and/or environment is recorded.
PASS FAIL	Results :
	END RECORDING OF POST-TEST INFORMATION
End of Test Procedure 4.3.3.7.1	



CHAPTER 5

Traceability

ALL-TRACEABILITY :: THIS SECTION SHALL TRACEABILITY BETWEEN THE DEFINING REQUIREMENTS AND THE TESTS PLANNED TO PERFORM THE VALIDATION OF THE REQUIREMENTS. THIS ARTIFACT MAPS:

- (A) EACH TEST IDENTIFIED IN THIS ARTIFACT TO THE UNDERLYING REQUIREMENTS AND, IF APPLICABLE, SYSTEM REQUIREMENTS IT ADDRESSES. (ALTERNATIVELY, THIS TRACEABILITY MAY BE PROVIDED ELSEWHERE IN THIS ARTIFACT AND REFERENCED FROM THIS PARAGRAPH.), OR
- (B) FROM EACH CAPABILITY REQUIREMENT AND, IF APPLICABLE, EACH SUB-SYSTEM REQUIREMENT COVERED BY THIS ARTIFACT TO THE ARTIFACT(S) THAT ADDRESS IT. THE TRACEABILITY SHALL COVER THE REQUIREMENTS IN ALL-APPLICABLE ARTIFACTS OR, IF APPLICABLE, THE SYSTEM DESIGN REQUIREMENTS IN THE **SSS**.

This section provides traceability of the system components and interfaces to the design requirements. In general, the requirements are included in the individual tests, but they can be summarized here via automatic tools that read the test procedure files and generate this summary.



CHAPTER 6

Notes

ALL-NOTES :: THIS SECTION SHALL CONTAIN ANY GENERAL INFORMATION THAT AIDS IN UNDERSTANDING THIS ARTIFACT (E.G., BACKGROUND INFORMATION, RATIONALE, ETC.)

This chapter is explanations about how test are setup and how we use them. The most basic test is used to to if the system turns on.

6.1 Note Area 1

ALL-NOTES :: THIS SECTION SHALL CONTAIN ANY GENERAL INFORMATION THAT AIDS IN UNDERSTANDING THIS ARTIFACT (E.G., BACKGROUND INFORMATION, RATIONALE, ETC.)

This section is to see how to test and move to the help screen.

6.2 Note Area 2

ALL-NOTES :: THIS SECTION SHALL CONTAIN ANY GENERAL INFORMATION THAT AIDS IN UNDERSTANDING THIS ARTIFACT (E.G., BACKGROUND INFORMATION, RATIONALE, ETC.)

This section is see if the system can start the game and see if it works.



APPENDIX

Additional Information

ALL-APPENDIX :: THIS SECTION SHALL CONTAIN ANY GENERAL INFORMATION THAT AIDS IN UNDERSTANDING THIS ARTIFACT (E.G., BACKGROUND INFORMATION, RATIONALE, ETC.)

This section provides additional information, as necessary, to augment the **STS**.



Index

All To Be Determined Items, 9

Classification Level
 Cntrl-Unclass-Info, 1

Glossary
 Customer, 5

MIL-STD-498
 DID, 3
 SSS, 44
 STD, 1, 3, 46
 STR, 3

Point of Contact, 1, 3

This System, 1

Yourdon's Structured Method, 5